

ASSESSMENT FILE : CompTia A+

AP-A1

Section A — Concept Application

SCENARIO You are setting up a new office workstation and need to choose the core hardware for it from a supplier's parts list.

Question: 1 Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.



1. Select the Motherboard

When selecting hardware for a new office workstation, compatibility between the components is very important. I would first select a suitable motherboard and then choose a compatible CPU, RAM, and storage device based on the motherboard specifications. I would select a motherboard according to the requirements of the office workstation. I would check its CPU socket, supported RAM type and capacity, RAM speed, SATA ports, M.2 slots, PCIe expansion slots, and form factor. The motherboard should also fit the selected computer case.

2. Select the CPU

The CPU must be compatible with the motherboard's CPU socket. I would check the motherboard manufacturer's CPU compatibility list to confirm that the selected processor is supported. I would also consider the CPU's performance and power requirements for office applications.

3. Select RAM

I would check which type of RAM is supported by the motherboard, along with the maximum supported capacity and speed. The selected RAM must match the motherboard's supported RAM specification. I would choose an appropriate capacity based on the requirements of office applications.

4. Select STORAGE

I would check whether the motherboard supports SATA storage, M.2 storage, or both. If using an NVMe SSD, I would confirm that the motherboard has a compatible M.2 slot that supports NVMe. For a SATA SSD or HDD, I would confirm that SATA ports are available. Before purchasing or assembling the system, I would compare the specifications of all components. I would verify the CPU socket and CPU support, RAM type and supported capacity, and the storage interface. I would also check the motherboard manufacturer's compatibility information. After assembly, I would enter the BIOS/UEFI and confirm that the CPU, RAM, and storage device are detected correctly.

Therefore, checking the specifications and compatibility information before assembly helps prevent hardware incompatibility problems and ensures that all selected components can work together correctly.

Motherboard = Main platform

Then check:

CPU → Socket

RAM → Type + Capacity + Speed

Storage → SATA/M.2 + Supported type

And finally:

BIOS/UEFI → Confirm CPU + RAM + Storage are detected.

SCENARIO

You are advising a client whose employees mainly run office and productivity applications on their workstations.

Question 2

Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.



HDD, SSD, and NVMe are storage technologies used to store operating systems, applications, documents, and other data. They differ mainly in their technology, performance, and interface.

HDD (Hard Disk Drive):

An HDD stores data on magnetic spinning disks and uses mechanical moving parts. It generally provides large storage capacity at a lower cost, but it is slower than SSD storage. Because it contains moving parts, it can also be more affected by physical shock and produces some mechanical noise.

SSD (Solid State Drive):

An SSD uses flash memory and does not contain moving mechanical parts. It provides faster boot times, faster application loading, silent operation, and better resistance to physical shock compared with an HDD. SSDs can use different interfaces, including SATA.

NVMe SSD:

NVMe is a storage protocol designed for SSDs that communicate through the PCIe interface. NVMe SSDs can provide higher performance and lower latency than SATA-based SSDs because they can use the higher bandwidth of PCIe. They are commonly found in M.2 form factors.

Recommended Option for Office Workstations:

For employees mainly using office and productivity applications, I would recommend an **SSD** because it provides faster system booting, quicker application loading, and better overall responsiveness than a traditional HDD. If the workstation and workload require higher storage performance, an NVMe SSD can be selected because it provides higher performance through the PCIe interface.

Therefore, an SSD is a suitable choice for modern office workstations because it provides a good combination of speed, reliability, and responsiveness for everyday productivity tasks.

HDD → Magnetic + Moving parts + Slower

SSD → Flash + No moving parts + Faster

NVMe → SSD + PCIe + Higher performance

Scenario

You are configuring a workstation that will run mildly graphics-intensive design software for several hours a day.

Question 3

Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.



A GPU (Graphics Processing Unit) is responsible for processing graphics-related tasks and generating the visual output displayed on the monitor. In a workstation running mildly graphics-intensive design software, the GPU can help process graphics, rendering, visual effects, and other graphical workloads. A suitable GPU can improve the responsiveness and performance of the design applications.

The cooling system is responsible for removing the heat generated by components such as the CPU and GPU during operation. It may include heatsinks, fans, case fans, and thermal compound. Good cooling keeps the components within their safe operating temperature range.

When the workstation is used for several hours continuously, the CPU and GPU may produce significant heat. If the heat is not removed effectively, the components may become too hot and reduce their performance through thermal throttling. Excessive temperatures can also cause system instability or unexpected shutdowns.

Therefore, a suitable GPU provides the required graphics-processing performance, while an effective cooling system helps maintain stable temperatures. Together, they allow the workstation to maintain reliable performance during sustained graphics-intensive workloads.

GPU = Processes graphics

Cooling = Removes heat

Too much heat → Thermal throttling → Lower performance

Graphics workload creates more processing demand → components generate heat → cooling removes heat → system maintains stable performance.

Scenario

You are assembling a PC from components sourced from three different vendors.

Question 4

Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components



Hardware compatibility is important when assembling a computer because all components must be able to communicate, connect, and operate correctly with each other. Components can be purchased from different vendors, but their technical specifications must be compatible.

For example, the CPU must be supported by the motherboard's CPU socket and CPU compatibility list. The RAM must match the type, capacity, and supported specifications of the motherboard. Similarly, the storage device must use a connection supported by the motherboard, such as SATA or a compatible M.2 interface. The PSU must also provide sufficient power and the required power connectors for the system.

If incompatible components are used, several problems can occur. The computer may fail to boot, a component may not be detected, the system may become unstable, or the hardware may not operate at its expected performance. Physical compatibility problems can also occur if a component does not fit the motherboard, expansion slot, or computer case.

Therefore, before assembling a computer, I would check the specifications and compatibility information for each component. This helps ensure that the components work together correctly and prevents installation, boot, power, and performance problems.

-  CPU must be supported by the motherboard.
-  RAM must match the motherboard's supported memory type.
-  Storage must use a supported connection.
-  PSU must provide the required power and connectors.
-  Components must physically fit in the case.

Scenario

You are finalizing the build and need to connect the power supply unit (PSU) to the motherboard, GPU, and storage devices before first boot.

Question 5

Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.



The PSU (Power Supply Unit) provides electrical power to the components of a computer. To select an appropriately rated PSU, I would consider the power requirements of the CPU, GPU, motherboard, storage devices, cooling fans, and other components. I would then select a PSU with sufficient wattage to safely support the complete system and suitable connectors for the required components.

Before powering on the newly assembled computer, I would perform several safety and connection checks. First, I would ensure that the PSU is switched off and disconnected from mains power while checking the internal connections. I would follow ESD precautions when handling the components and make sure there are no loose screws or other objects inside the case.

I would check that the motherboard is correctly installed and that the CPU, RAM, GPU, and storage devices are properly seated and connected. I would also confirm that the CPU cooler is installed correctly.

For power connections, I would verify that the main motherboard power connector is connected correctly, the CPU power connector is attached, the required PCIe power connector is connected to the GPU if necessary, and SATA power connectors are connected to SATA storage devices when used.

Finally, I would inspect all connections for loose or incorrect cables and make sure no connector has been forced into the wrong socket. After completing these checks, I would connect the PSU to mains power and perform the first boot.

Think **PSU = Power for everything**.

Before first boot:

1. **Correct wattage** ⚡
2. **Correct connectors** 🔌
3. **Components seated properly** □
4. **ESD/safety checked**
5. **No loose screws/cables**

SCENARIO

You are handing over a freshly assembled workstation to the client and need to confirm it is ready for use.

QUESTION 6

Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.



POST (Power-On Self-Test) is an initial hardware check performed by a computer when it is powered on. After assembling a workstation, I would press the power button and check that the system powers on correctly, the fans operate, and the display becomes active.

I would observe whether the computer completes POST without displaying any hardware error. I would also check for beep codes, diagnostic LEDs, or POST codes if they are provided by the motherboard. After POST, I would enter the BIOS/UEFI and confirm that the CPU, RAM, and storage device are detected correctly.

If the system does not pass POST, I would troubleshoot the hardware systematically rather than replacing components randomly. First, I would check the PSU and all power connections. Next, I would check that the RAM is correctly seated. I would then check the CPU installation and CPU power connection, followed by the GPU and display connections if applicable. I would also check the storage connections and verify whether the storage device is detected in BIOS/UEFI.

I would use any available beep codes, diagnostic LEDs, or POST codes to identify the possible source of the problem. I would also verify motherboard and CPU compatibility if necessary.

Therefore, a successful POST together with correct hardware detection in BIOS/UEFI confirms that the newly assembled workstation has successfully completed its initial hardware startup checks.

POST = Initial hardware check

After pressing Power:

Power ON → POST → BIOS/UEFI → Hardware detected → Operating System

If POST fails:

PSU → Connections → RAM → CPU → GPU/Display → Storage

The key point is **don't randomly replace parts**. Check the likely hardware causes systematically.

Section B — Practical Coding Tasks

TASK 1

Component Identification & Functions



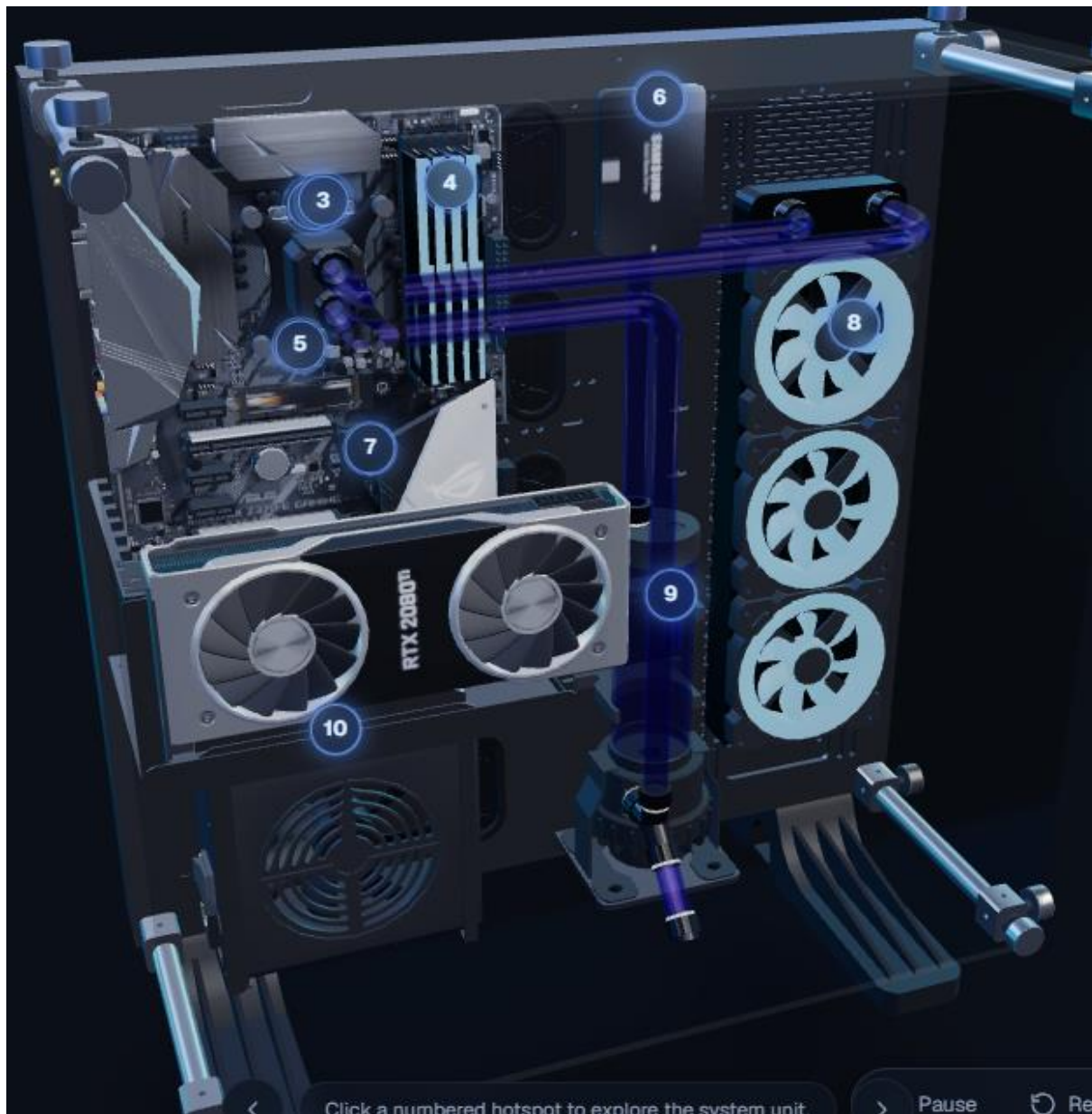
Task 1: Component Identification & Functions

The following are the core hardware components identified in the desktop system:

No.	Component	Function	Make/Model or Key Specification
1	Motherboard	The motherboard is the main circuit board of the computer. It connects and allows communication between the CPU, RAM, storage, GPU, PSU, and other hardware components.	To be filled from the physical hardware
2	CPU	The CPU (Central Processing Unit) executes instructions and performs calculations required by the operating system and applications.	To be filled from the physical hardware
3	RAM	RAM (Random Access Memory) temporarily stores data and programs that are currently being used by the CPU. It is volatile memory, so its contents are lost when the computer is powered off.	To be filled from the physical hardware
4	GPU	The GPU (Graphics Processing Unit) processes graphics and helps generate the visual output displayed on the monitor. It may be integrated or dedicated.	To be filled from the physical hardware
5	Storage Device	The storage device provides long-term storage for the operating system, applications, documents, and other files. Examples include HDD, SATA SSD, and NVMe SSD.	To be filled from the physical hardware
6	Cooling System	The cooling system removes heat generated by components such as the CPU and GPU. It may include heatsinks, fans, and thermal compound.	To be filled from the physical hardware

Summary

Each component has a different role in a desktop computer. The motherboard provides the connections, the CPU processes instructions, RAM provides temporary working memory, the GPU handles graphics processing, storage keeps data permanently, and the cooling system helps maintain safe operating temperatures.



- 3 → CPU
- 4 → RAM
- 5/7 → Motherboard
- 6 → Storage
- 8/9 → Cooling
- 10 → GPU (RTX 2080 Ti)

Hotspot	Component	Function	What we can identify
3	CPU	Processes instructions and performs calculations for the computer.	CPU is located under the CPU cooling block. Exact model is not visible.
4	RAM	Temporarily stores data and programs currently being used by the CPU.	Multiple RAM modules are installed. Exact capacity/model is not visible.
5 / 7	Motherboard	Main circuit board that connects the CPU, RAM, storage, GPU and other components.	Motherboard is clearly visible. Exact model is not readable.
10	GPU	Processes graphics and sends visual output to the monitor.	RTX 2080 Ti is clearly written on the graphics card.
6	Storage device	Permanently stores the operating system, applications and user files.	A Samsung storage device is visible; exact model/capacity is not readable.
8 / 9	Cooling system	Removes heat from components and maintains safe operating temperatures.	Case fans and liquid-cooling components are visible.

TASK 2

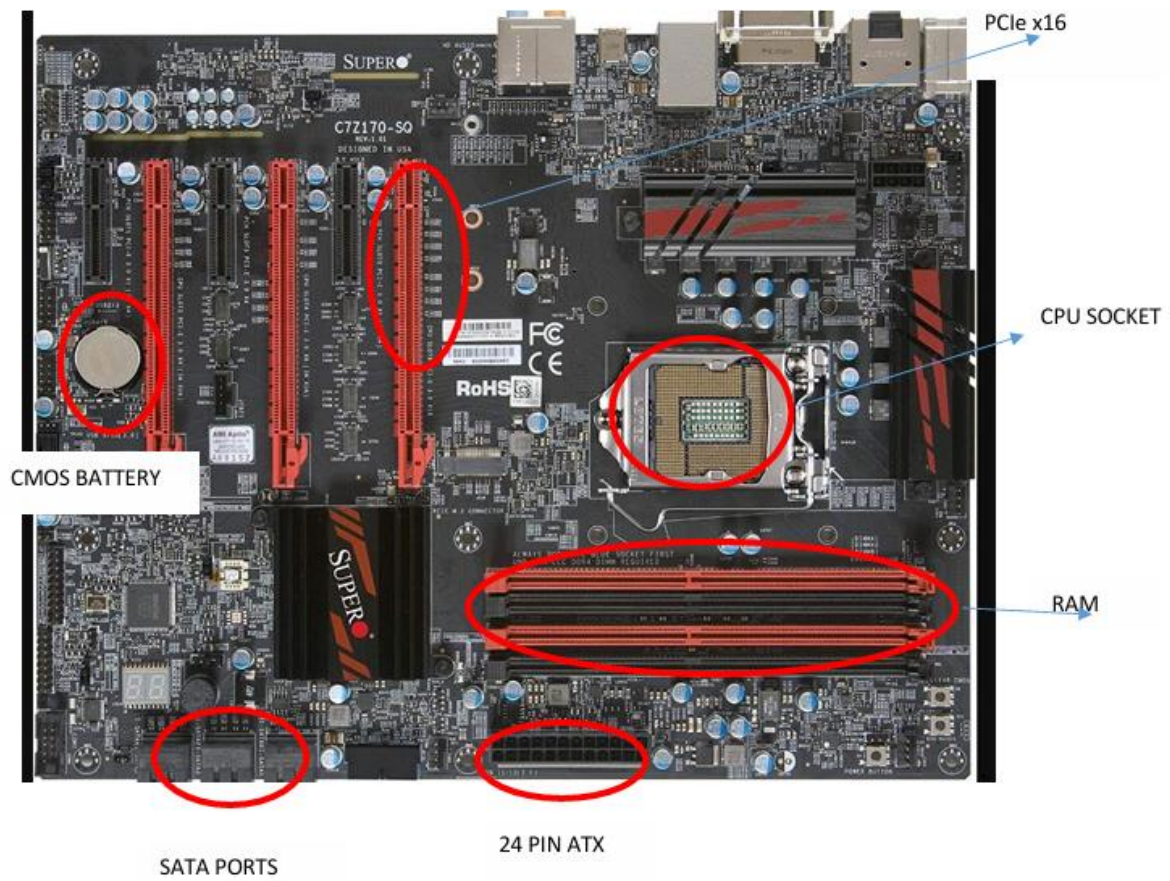
Motherboard Ports & Socket Identification



The motherboard is the main circuit board of the computer and contains the CPU socket, memory slots, expansion slots, storage interfaces, power connectors, and internal headers.

No.	Motherboard Component	Purpose
1	CPU Socket	Holds the CPU and provides the electrical connection between the CPU and motherboard.
2	DIMM/RAM Slots	Used to install RAM modules for temporary storage of actively used data and instructions.
3	PCIe Expansion Slots	Used to install expansion cards such as graphics cards, network cards, and sound cards.
4	M.2 Slot	Used for compatible M.2 storage devices, including NVMe SSDs.
5	SATA Ports	Used to connect SATA storage devices such as HDDs and SATA SSDs.
6	24-pin ATX Power Connector	Provides the main power connection from the PSU to the motherboard.
7	CPU Power Connector	Provides dedicated power to the CPU through the motherboard.
8	USB Headers	Connect the computer case's internal/front USB ports to the motherboard.
9	Front-Panel Headers	Connect the power button, reset button, LEDs, and other case controls to the motherboard.
10	Rear I/O Ports	Provide external connections such as USB, network, audio, and display connections.

CPU Socket → RAM/DIMM Slots → PCIe Slots → M.2 → SATA → 24-pin Power → CPU Power → USB Headers → Front-Panel Headers



Task 3: RAM & Storage Installation

RAM Installation

First, I would shut down the computer, switch off the PSU, disconnect the power source, and follow appropriate ESD precautions. I would locate the RAM module in the motherboard's DIMM slot and release the retaining clips. I would carefully remove the RAM module by holding it from its edges and avoiding unnecessary contact with the gold contacts.

To reinstall the RAM, I would align the notch on the RAM module with the key in the DIMM slot. I would press the module down evenly until it is fully seated and the retaining clips lock into position. I would visually check that the RAM is correctly aligned and securely installed.

Storage Installation

I would first identify whether the storage device uses **SATA or NVMe/M.2**.

For a SATA storage device, I would disconnect the SATA data and power connections, remove the drive, reinstall it, and reconnect both cables securely.

For an M.2/NVMe SSD, I would remove its retaining screw, carefully remove the SSD, insert it into the M.2 slot at the correct angle, press it down, and secure it with the retaining screw.

Verification

After reinstalling the RAM and storage device, I would power on the computer and enter the BIOS/UEFI. I would verify that the correct amount of RAM is detected and that the storage device appears in the storage/device information.

If both components are detected correctly, I would confirm that the RAM and storage installation was successful. If either component is not detected, I would power off the system and recheck the seating, connections, and compatibility.

Component	Action	Connection	Verification
RAM	Removed and reinstalled	DIMM slot	RAM detected in BIOS/UEFI
Storage	Removed and reinstalled	SATA / M.2	Storage detected in BIOS/UEFI



Figure 3: RAM and Storage Installation Procedure

★ Remember

RAM:

Remove → Align notch → Press firmly → Clips lock → BIOS detects RAM

SATA storage:

Remove → SATA data + power → Reinstall → Reconnect → BIOS detects drive

NVMe:

Remove screw → Insert into M.2 → Secure → BIOS detects drive

Task 4: Full Desktop PC Assembly

I assembled the desktop PC by following a systematic installation sequence and observing appropriate ESD and safety precautions.

1. Workspace Preparation:

I prepared a clean working area and followed ESD precautions. The PSU was kept switched off and disconnected from power while internal components were being installed.

2. CPU Installation:

I installed the CPU into the compatible motherboard CPU socket by aligning it correctly with the socket markings. I secured the CPU without applying unnecessary force.

3. Cooling System Installation:

I installed the CPU cooling system and ensured that the cooler was securely mounted. The required CPU cooling connection was connected to the appropriate motherboard header.

4. RAM Installation:

I installed the RAM modules into the motherboard's DIMM slots. I aligned the notch on each module with the slot and pressed the modules into place until the retaining clips secured them.

5. Storage Installation:

I installed the storage device using the appropriate connection. For an M.2/NVMe device, it was installed into the compatible M.2 slot and secured. For SATA storage, the SATA data and power connections were connected correctly.

6. Motherboard Installation:

I positioned the motherboard correctly inside the case and secured it using the appropriate mounting points.

7. PSU Installation and Connections:

I installed the PSU and connected the required power cables, including the main motherboard power connector and CPU power connector. The required GPU and storage power connections were also connected.

8. GPU Installation:

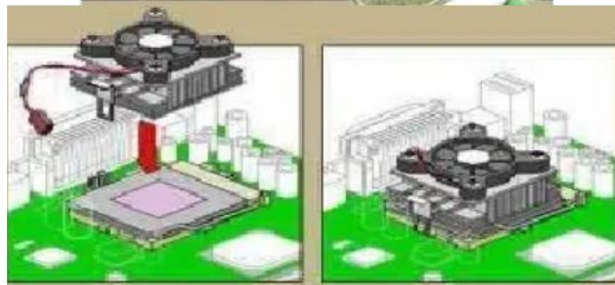
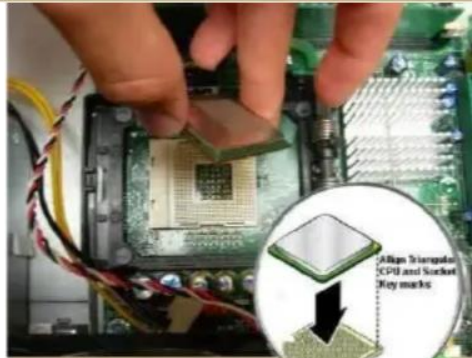
I installed the graphics card into the appropriate PCIe expansion slot and secured it. The GPU's required power connection was also connected.

9. Final Inspection:

Before powering on the system, I checked that the CPU, cooler, RAM, storage, GPU, PSU cables, and other required connections were installed correctly. I also checked that there were no loose screws or objects inside the case.

10. Power-On and POST Verification:

I powered on the computer and checked that the system started correctly and completed POST. I then accessed the BIOS/UEFI and verified that the CPU, RAM, and storage device were detected correctly.



Section C — Mini Capstone Project

Mini Project: Workstation Deployment & Boot-Failure Diagnosis

OBJECTIVE

ABC Solutions is setting up a new computer lab. After assembling several systems, some computers fail to start while others display hardware detection errors, and the IT team suspects incorrect installation or compatibility issues. Build (or inspect) a workstation, diagnose the likely causes of these failures, and produce a diagnostic and installation-verification report — combining component selection, installation procedure, and compatibility troubleshooting from Section B.

1. Three hardware components to check first when a newly assembled PC fails to boot



If a newly assembled PC fails to boot, I would first check the **PSU/power connections, RAM, and CPU/motherboard**.

1. PSU and Power Connections:

I would check that the PSU is switched on and connected correctly. I would verify the 24-pin motherboard power connector, CPU power connector, and required GPU power connection. Incorrect or missing power connections can prevent the system from starting.

2. RAM:

I would check that the RAM modules are installed in the correct DIMM slots and are fully seated. Incorrectly installed RAM can cause POST failure and hardware-detection problems.

3. CPU and Motherboard:

I would check that the CPU is installed correctly, the CPU power connection is connected, and the CPU is compatible with the motherboard. CPU–motherboard compatibility is necessary for the system to initialize correctly.

These checks cover important power, memory, and processing components that can prevent a newly assembled system from starting.



Motherboard–CPU Compatibility

Motherboard–CPU compatibility is important because the CPU must be supported by the motherboard for the system to initialize and operate correctly. I would check the CPU socket type and the motherboard manufacturer's supported CPU list before installation.

In the workstation I inspected, I would verify that the installed CPU matches the motherboard's supported socket and CPU compatibility information. I would also ensure that the required CPU power connection is properly connected.

If an incompatible CPU is installed, the computer may fail to boot or complete POST, even if the CPU physically fits the motherboard. Therefore, checking compatibility before assembly helps prevent boot failures and hardware-detection problems.



RAM and Storage Installation Verification

To verify the RAM installation, I would first inspect the RAM modules and confirm that they are correctly aligned and fully seated in the DIMM slots. The retaining clips should be securely locked. After powering on the system, I would enter BIOS/UEFI and confirm that the expected RAM capacity is detected.

For storage, I would check that the storage device is correctly installed and that all required connections are secure. For SATA storage, I would check both the SATA data and power connections. For an M.2/NVMe SSD, I would check that it is correctly inserted into the M.2 slot and secured.

After powering on the system, I would enter BIOS/UEFI and verify that the storage device is detected. Successful POST and correct detection of RAM and storage provide evidence that these components are installed correctly.



Hardware Installation checklist

No.	Checklist Item	Purpose
1	Verify CPU and motherboard compatibility before installation.	Prevents incompatible processor and motherboard problems.
2	Follow ESD precautions while handling components.	Reduces the risk of electrostatic damage.
3	Check that RAM, CPU, GPU, and storage are properly seated.	Prevents connection and hardware-detection problems.
4	Verify all required PSU power connections before first boot.	Prevents power-related startup failures.
5	Enter BIOS/UEFI after assembly and verify CPU, RAM, and storage detection.	Confirms that important hardware has been recognized correctly.



Technical Justification for Using SSDs

I would use SSDs instead of traditional HDDs in modern office workstations because SSDs use flash memory and do not contain mechanical moving parts. This allows them to provide faster system booting, faster application loading, and better overall system responsiveness.

For the ABC Solutions computer lab, employees are expected to use modern office and productivity applications. Faster storage can reduce waiting time when starting the operating system, opening applications, and accessing files. SSDs also operate silently and are less affected by physical shock than mechanical HDDs.

Therefore, SSD storage is appropriate for the lab's modern office workstations because it provides responsive storage performance for everyday productivity tasks.

Section D — AI-Augmented Learning

STEP 1 · BUILD WITH AI

Use an AI tool of your choice (ChatGPT, Claude, Copilot, etc.) to help you produce a step-by-step troubleshooting checklist that:

STEP 2 · TEST & DEBUG (WITHOUT AI)

Then, working without AI, review the checklist and find at least one step that is missing, incorrectly ordered, or unsafe (for example, skipping ESD precautions or checking software before physical connections). Fix it yourself.



Exact Prompt Used

"Create a clear step-by-step troubleshooting checklist for a junior technician diagnosing a desktop PC that fails to boot immediately after assembly. Put the steps in the correct priority order, starting with basic physical and power checks before more detailed checks. Include the motherboard, RAM, storage, and PSU as possible failure points. Make the checklist simple enough to follow on-site. Apply the checklist to the ABC Solutions scenario from the assessment, where newly assembled computers fail to start or show hardware detection errors, and identify which troubleshooting step would have caught the reported issue."



Desktop PC Boot-Failure Troubleshooting Checklist

Step 1 — Check the PSU and power

- Confirm the PSU is switched on.
- Check the power cable and wall connection.
- Check the 24-pin motherboard power connector.
- Check the CPU power connector.
- Check the required GPU power connector.

Step 2 — Perform a physical inspection

- Check that there are no loose screws or objects inside the case.
- Check that components are correctly installed.
- Check that cables are not loose or incorrectly connected.

Step 3 — Check the RAM

- Confirm the RAM is fully seated.
- Check that the RAM is installed in the appropriate slot.
- If necessary, remove and reinstall the RAM.

Step 4 — Check the motherboard and CPU

- Check the CPU installation.
- Check the CPU power connection.
- Check motherboard–CPU compatibility.
- Look for motherboard diagnostic LEDs or POST codes.

Step 5 — Check the GPU/display

- Confirm the GPU is correctly seated.
- Check required GPU power connections.
- Check the display cable and monitor input.

Step 6 — Check storage

- Check SATA data and power connections if using SATA storage.
- Check that an M.2/NVMe SSD is correctly installed.
- Enter BIOS/UEFI and check whether the storage device is detected.

Step 7 — Check BIOS/UEFI

- Enter BIOS/UEFI.
- Confirm that the CPU and RAM are detected.
- Confirm that the storage device is detected.
- Check for hardware or configuration errors.

Step 8 — Check POST indications

- Observe beep codes.
- Check diagnostic LEDs or POST codes if available.
- Use these indications to identify the possible failing component.

Application to ABC Solutions

For ABC Solutions, the first checks should focus on the physical installation and power connections because the computers had just been assembled. If a power connection or incorrectly seated RAM caused the failure, the problem would have been identified during the PSU/power or RAM checks.

Step 3 — Test & Debug WITHOUT AI

Now comes the important part.

The assessment specifically tells **you** to review the AI checklist and find something that is **missing, incorrectly ordered, or unsafe**, then fix it yourself.

 **My review**

One improvement we can make is to explicitly place **ESD/safety precautions before physical component handling**.

The original checklist starts with checking power and then moves into physical inspection, but it doesn't clearly say:

Before opening the case or reseating components, shut down/disconnect power and follow ESD precautions.

That's an important safety step when physically handling RAM, CPU, GPU, or other components.

So we'll correct the checklist.

Step 4 — Corrected Checklist

Desktop PC Boot-Failure Troubleshooting Checklist — Corrected Version

Step 1 — Follow safety and ESD precautions

- Switch off the PSU.
- Disconnect the system from mains power before opening the case.
- Follow appropriate ESD precautions before handling components.

Step 2 — Check external power

- Check the wall power connection.
- Check the power cable.
- Confirm the PSU switch is in the correct position.

Step 3 — Check internal PSU connections

- Check the 24-pin motherboard power connector.
- Check the CPU power connector.
- Check the required GPU power connector.
- Check SATA power connections where applicable.

Step 4 — Perform a physical inspection

- Check for loose cables.
- Check for loose screws or objects inside the case.
- Check that components are installed correctly.

Step 5 — Check RAM

- Check that RAM is correctly aligned and fully seated.
- Check the appropriate DIMM slot.
- Reseat the RAM if necessary.

Step 6 — Check CPU and motherboard

- Check CPU installation.
- Check CPU power connection.
- Verify motherboard–CPU compatibility.
- Check motherboard diagnostic LEDs or POST codes if available.

Step 7 — Check GPU and display

- Check GPU seating.
- Check GPU power connections.
- Check display cable and monitor input.

Step 8 — Check POST/BIOS

- Power on the system.
- Observe POST behavior.
- Check beep codes, diagnostic LEDs, or POST codes.
- Enter BIOS/UEFI and verify CPU and RAM detection.

Step 9 — Check storage

- Verify SATA data/power connections or M.2 installation.
- Check whether the storage device is detected in BIOS/UEFI.

Step 10 — Apply the findings to ABC Solutions

For the ABC Solutions scenario, I would use the checklist from the beginning and identify the failure based on the observed symptoms. If the problem was caused by a loose power connection, the issue would be identified during the power-connection checks. If the RAM was incorrectly installed, the problem would be identified during the RAM inspection and reseating step.

Conclusion

I reviewed the AI-generated troubleshooting checklist and found that ESD and power-isolation precautions were not clearly placed before physical component handling. I added a dedicated safety step at the beginning of the corrected checklist. I also separated external power checks from internal component checks to make the troubleshooting sequence safer and easier for a junior technician to follow.